Scalability & Future Plan

Date	01 October 2026
Team ID	SWTID-2026-2430
Project Name	Team Enrollment Confirmation - FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	2 Marks

Scalability & Future Plan:

This document outlines how the current FitBuddy solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current prototype state, along with individual team work assignments.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-instance SQLite local storage and sequential execution.	Limits concurrent active sessions and may cause slow response times under heavy user load.	**High**
2	Synchronous API calls to Google Gemini AI service.	Increases request latency for generating 7-day plans during peak network congestion.	**Medium**
3	Basic user session management without persistent multi-device login.	Users cannot access their workout history dynamically across different devices.	**Low**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Local FastAPI server handling light single-user traffic.	Deploy with Gunicorn/Uvicorn workers on AWS ECS/Kubernetes with load balancers for auto-scaling.
2	**Data Storage**	File-based SQLite database for plan history.	Migrate to managed PostgreSQL / AWS RDS with connection pooling for high-concurrency read/write.
3	**Performance**	Synchronous generation of plans and recommendations.	Implement Redis caching for static prompts/plans and Celery asynchronous background task queues.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	**Security**	Basic input validation via Pydantic forms.	Add JWT-based OAuth2 authentication, HTTPS encryption, dynamic rate limiting, and API key secret rotation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core Gemini AI integration, FastAPI backend, dynamic Jinja2 UI, and SQLite local storage.	Completed	Functional prototype for generating custom 7-day fitness & recovery plans.
Phase 2	User Authentication, User Profiles, and Wearable Device (Fitbit/Apple Health) Integration.	Q1 2027	Personalized tracking using real-time heart rate and step activity data.
Phase 3	Multi-modal Computer Vision exercise form analysis and AI voice coach guidance.	Q2 2027	Reduces workout injury risk with real-time video form feedback.
Phase 4	Enterprise multi-tenant deployment, mobile apps (React Native), and community fitness leaderboards.	Q3 2027	Expands user engagement and enables commercial gym platform partnerships.

Team Work Allocation:

Work responsibilities assigned in exact order of member names provided:

S.No	Team Member	Assigned Responsibility	Expected Contribution
1	**Divya.D**	Requirements and Coordination	Define user inputs, fitness goals, workflows, and requirements; coordinate and consolidate documentation.
2	**Deepika.M**	Gemini AI Integration & Prompts	Design prompts for structured 7-day workout plans, nutrition/recovery tips, and feedback-based revisions.
3	**Dharani.T**	FastAPI Backend & Routing	Implement form handling and routes for plan generation and feedback; connect routes to AI functions.
4	**Divya shree.R**	Frontend & User Experience	Build HTML/Jinja2 pages for inputs, plans, tips, and feedback; style responsive layouts.
5	**DivyaDharshini.V**	Database, Testing & Deployment Support	Support SQLite/SQLAlchemy plan history, test main workflows, and verify local deployment and API documentation.

Shared Team Responsibility: All members review the final output, check that plans are clearly presented, and prepare the project demonstration.
Safety Note: Fitness suggestions generated by AI models are general information and should be adapted to individual user needs and health conditions.